

Design System & UX

polycarpic — personal budgeting & wealth management (virtual Single Family Office)

Owner: ux-designer agent **Status:** Draft — Plan phase, matured alongside ARCH **Version:** 0.4

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1. Overview & Design Principles

What this product should feel like to use, and the principles that resolve design disputes.

- **Mobile-first when in doubt.** The primary v1 target is a self-hosted web dashboard used on desktop for deep review (statements, lot tables) and on mobile for quick checks (dashboard glance, confirming a draft transaction) — design the dense screens up from a mobile layout, not down from desktop.
- **Accessible by default (WCAG AA).** Non-negotiable per PRD §6. Financial gain/loss and status signals never rely on color alone — see [Financial semantics](#).
- **No decorative loading states.** Skeletons match the real layout; spinners are a last resort (see [Component inventory](#)).
- **Every number earns trust.** This is a general ledger, not a budgeting toy — a balance, a delta, or a pending-liability figure must always be traceable to its source (the journal entry, the lot, the reconciliation). The UI never shows a number it can't explain if clicked.
- **Immutability is visible, not just enforced.** A posted transaction reads as posted (badge, disabled edit affordance); a correction reads as a reversal + new entry, never as a silent edit (PRD §5).
- **Diverge from polycarpic's own tooling, deliberately.** The cairn tracker (our internal tool) and the product are not the same thing and shouldn't look like it — this system's identity is derived from **measured** screenshots and live style reads off Monarch Money and Origin (`useorigin.com`), not memory (Principal direction, 2026-09-28) — see [Inspiration: Monarch & Origin](#).

2. Design System

The single source of truth for the visual language. The written spec lives alongside this doc; the machine-readable system lives in the two CSS files that `frontend-lead` consumes directly.

- [design-system-spec.md § Inspiration](#) — what was taken from Monarch Money and Origin, and why
- [design-system-spec.md](#) — the written spec (token rationale, financial semantics, component inventory, accessibility)
- `tokens.css` — design tokens as CSS custom properties (hand-authored palette, not a vendored shadcn preset)
- `screen.css` — component & screen styles built on the tokens

Figma stays the high-fidelity design tool for anything shipping pixel-exact (via the `ux-designer`'s Figma skills); `docs/DESIGN/` is the in-repo, code-level home that ships with the project and is the contract for implementation.

3. Design Tokens

Color, typography, spacing, radius, and shadow primitives — SvelteKit + shadcn-svelte (PRD §8). Edit values in [tokens.css](#); the swatches below preview the live variables. Full rationale in [design-system-spec.md](#).

3.1 Surfaces & Text

--background

--foreground

--card

--popover

--muted

--muted-foreground

3.2 Actions & States

--primary

--secondary

--accent

--destructive

3.3 Financial semantics (new — polycarpic-specific)

--positive

--warning

--destructive
(reused for negative/danger)

See [design-system-spec.md § Financial semantics](#) — gain/loss, safe-harbor status tiers, and the "never color alone" rule.

3.4 Borders & Focus

--border

--input

--ring

3.5 Charts (sequential ramp — magnitude, not category)

--chart-1

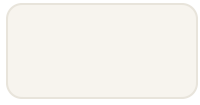
--chart-2

--chart-3

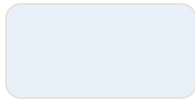
--chart-4

--chart-5

3.6 Sidebar



`--sidebar`



`--sidebar-accent`

Type: `--font-sans` *Inter* — one grotesque sans for body and headings, hierarchy by weight, confirmed against both reference products' actual app UI (their serif accent is marketing-only, see *Inspiration*); `--font-mono` *Roboto Mono* — Origin's own measured choice, for uppercase eyebrow labels, account numbers, tickers, tabular figures. **Radius:** measured `--radius-sm|md|lg|xl` = 6/8/12/20px (cards 12px, buttons 8px — both products' actual app-level values, not either's marketing-only pill).

4. Components & Screen Styles

The component inventory and the styles that realize it, built on the tokens. Styles live in [screen.css](#); full states-per-component table lives in [design-system-spec.md § Component inventory](#).

Component	shadcn-svelte primitive	polycarpic usage
Net-worth hero	Chart (LayerChart, Area)	Big figure + signed delta on the figure + single-line area chart — direct Monarch pattern
Grouped account list	Table / custom	Type-grouped rows with sparkline + freshness — direct Monarch pattern
Segmented bar + legend	Custom, tokens-driven	Assets/Liabilities, allocation actual-vs-target — direct Monarch pattern
Badge	Badge	Entity/account/transaction status — see Financial semantics
Transaction table	Table	Ledger, import staging, exceptions report
Entity switcher	Tabs / Sidebar	Switch between Person/Trust/Business/Household
Transaction detail	Sheet	Drill into a posted/draft entry, its journal lines, source doc
Empty state	Card + plain text, no skeleton	Dashboard panels not yet shipped (tax at 1.0, allocation at M4) — see styled-screens/dashboard.html

5. User Flows

How users move through the product to accomplish the highest-traffic stories (PRD §4). Source diagrams in [flows/](#).

- [Dashboard glance](#) — per-entity financial health at a glance
- [Import & confirm transactions](#) — staged drafts → reviewed → posted
- [Tax liability & safe-harbor](#) — pending liability, required payment, gap, pay
- [Securities lots & balance sheet](#) — GL vs. custodial reconciliation
- [Entity switch & membership access](#) — owner/manager/viewer boundary

6. Wireframes

Low-fidelity layout of each key screen — structure before style. Assets in [wireframes/](#).

- [Dashboard](#) — per-entity financial health
- [Ledger / Transactions](#) — import staging + confirm
- [Accounts](#) — GL vs. custodial balance sheet
- [Entities / Memberships](#)
- [Securities / Lots](#) — lot table + allocation view
- [Import](#) — provider link + batch review

7. Styled Screens

High-fidelity screens that apply the design system to the wireframes, consuming `tokens.css` + `screen.css` — a visual regression reference and a head-start for `frontend-lead`. Prioritized by traffic (PRD §4 stories 1 & 2); remaining screens promote from wireframe fidelity as the design system stabilizes.

- [Dashboard](#)
- [Ledger / Transactions](#)

8. Accessibility

Accessibility is a design responsibility, not a QA afterthought. Full table in [design-system-spec.md § Accessibility](#).

Concern	Standard / approach
Color contrast	WCAG AA (4.5:1 text, 3:1 large/UI) — every token pair measured, both modes, against this palette's actual hex values (chart ramp re-tuned after a first pass failed the 3:1 floor at both extremes); no CVD simulation run yet (open item)
Color-alone signaling	Never — every gain/loss/status pairs color with a sign/icon/label
Keyboard	All interactive elements reachable & operable (shadcn default)
Screen readers	Semantic markup; labelled controls; delta sign readable from text content
Motion	Respect <code>prefers-reduced-motion</code> (policy not yet written)

9. Open Questions

Unresolved design decisions and assumptions that could be wrong. Batched to team-lead for relay to the Principal — not blocking.

- **Resolved 2026-09-28:** visual identity diverges from cairn's tooling palette, and is now derived from measured Monarch/Origin references rather than general recall — see [design-system-spec.md § Inspiration for the correction history](#) (two prior passes rejected).
- No CVD (color-vision-deficiency) simulation run on the new palette yet — recommended before lock ([design-system-spec.md § Accessibility](#)).
- The Accounts screen's full grouped-account-list + segmented-bar treatment was only built on the Dashboard styled screen this pass (the highest-traffic comparison point) — Accounts itself is still at wireframe fidelity; promote it the same way as a fast-follow if the gap matters before Implement.
- Native iOS/macOS token export format (kickoff §2.10) — not yet decided, non-blocking for web.
- `prototype/` throwaway UX prototype (kickoff §2.9) not yet built — confirm ownership/timing with team-lead.

10. Appendix

- PRD user stories & functional requirements: [docs/PRD/index.html §4–5](#)
- Kickoff record, frontend/design system decisions: [docs/project_kickoff.md §2.9](#)

10.1 FAQs

Q: How do I add review comments to this design doc?

A: Hand-edit `docs/DESIGN/comments.md` with `## §<section-id>` blocks, or run `/serve-docs DESIGN` and use the inline `+ Comment` widget. Then run `/refine-doc DESIGN`. See [process/WORKFLOW.md](#) → Doc review loop.

Q: Why is the design system in `docs/DESIGN/` and not in the app's source?

A: It's the design *deliverable* — the contract. During Implement, `frontend-lead` imports or derives the app's real stylesheets from `tokens.css / screen.css`.

Q: Where did `variants.css` go?

A: It was generated infrastructure for the `cairn` tracker tool's own theme switcher, never part of the product — removed from this directory (POLY-78 relocates its generator target the same way). See [design-system-spec.md § Provenance & shared tooling](#).